

Course Name : Physics II Lab

Course Code : PHY 4242

Experiment no. : 03

Experiment Name : Verification of KVL

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Section : 1 (Lab group 1B)

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Objective:

This experiment is intended to verify Kirchhoff's Voltage Law (KVL) with help of series-parallel circuits.

Theory:

When current flows through a closed loop certain voltage drops occur for components.

According to Kirchhoff's Voltage Law,

"Around any closed circuit, the algebraic sum of voltage rises equals the algebraic sum of the voltage drops."

KVL is very useful for knowing unknown voltages and solving circuit related problems.

It is important to pay attention to the algebraic signs of the voltage drops across the elements.

Circuit Diagram:

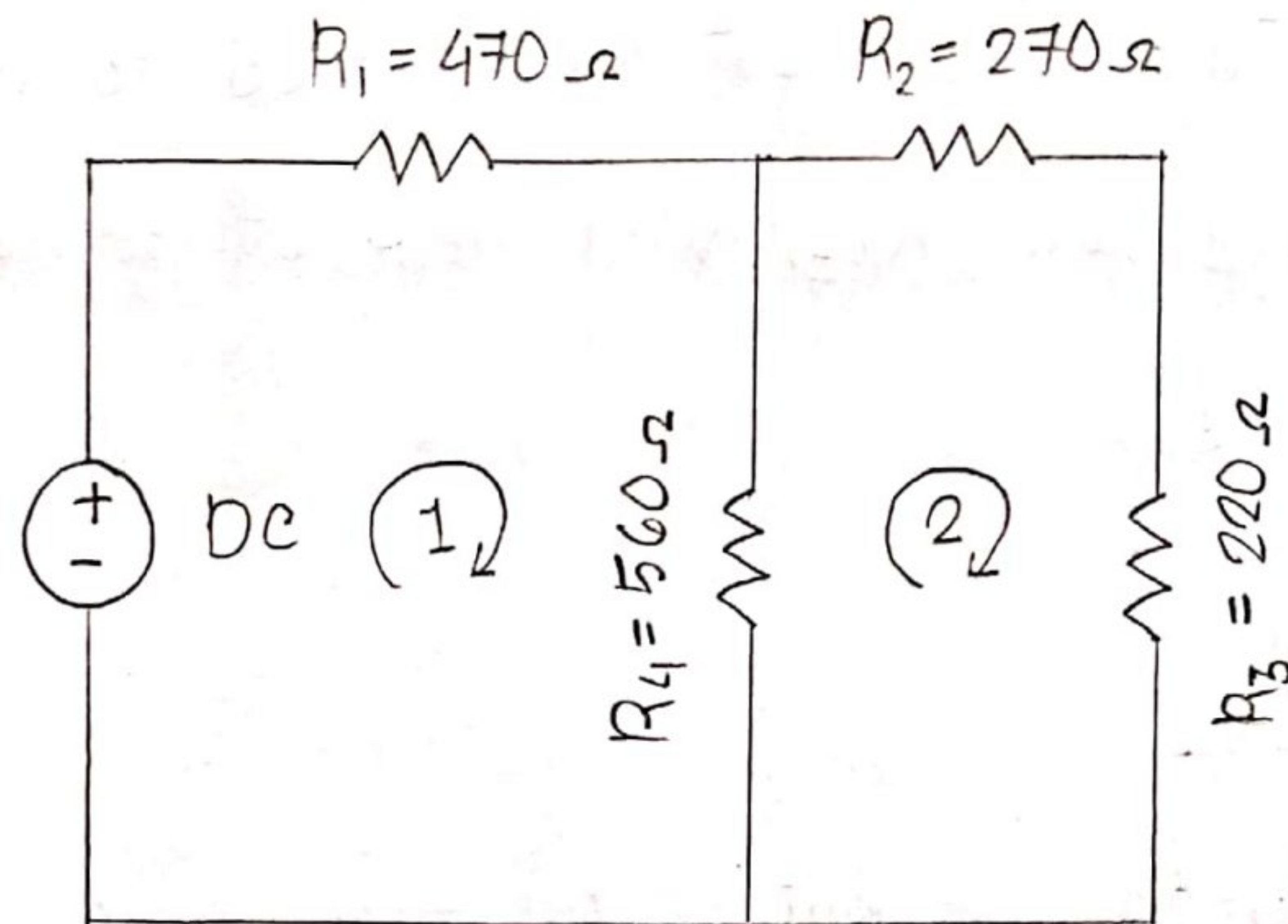


Figure: Circuit diagram of the experiment

Data Collection:

No. of Obs.	Source Voltage (V)	Voltage across (V)			
		R_1	R_2	R_3	R_4
1	5.07	3.231	0.993	0.821	1.815

Data Analysis:

We get, amount of resistors,

$$R_1 = 460\ \Omega$$

$$R_2 = 269.7\ \Omega$$

$$R_3 = 223\ \Omega$$

$$R_4 = 543\ \Omega$$

$$\text{Equivalent resistance, } R_{eq} = R_1 + \{(R_2 + R_3)^{-1} + R_4^{-1}\}^{-1}$$

$$= 718.314 \Omega$$

By taking the direction of current flow as clockwise,

In loop 1,

$$\text{DC voltage, } V = \cancel{5.046 \text{ V}} \quad 5.07 \text{ V}$$

$$V_{R_1} = 3.231 \text{ V}$$

$$V_{R_4} = 1.815 \text{ V}$$

$$\therefore V_{R_1} + V_{R_4} = (3.231 + 1.815) \text{ V}$$

$$= \cancel{5.046 \text{ V}} \quad 5.07 \text{ V}$$

$$\approx V$$

In loop 2,

$$V_{R_2} = 0.993 \text{ V}$$

$$V_{R_3} = 0.821 \text{ V}$$

$$V_{R_4} = 1.815 \text{ V}$$

$$\therefore V_{R_2} + V_{R_3} = (0.993 + 0.821) \text{ V}$$

$$= 1.814 \text{ V}$$

$$\approx V_{R_4}$$

Results :

By taking clockwise flow of current we can see in loop 1,

$$V_{R_1} + V_{R_4} \approx V$$

and in loop 2,

$$V_{R_2} + V_{R_3} \approx V_{R_4}$$

Thus KVL is verified.

Discussion and Q&A:

By performing the experiment KVL has been verified. A little discrepancy was found in the 1st and second loop. This is due to human error while measuring.

Yes, KVL can be applied to open circuit, because the voltage drop across the open part of the circuit is numerically equal to the supply voltage and hence the algebraic sum of the voltage drops around that particular loop is zero.

Some things that need to be taken into account while performing the experiment -

- i) Paying attention to the signs of the voltage across the elements.
- ii) The elements in the circuit were connected properly so that no short circuit occurred.

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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

COURSE: EEE 4242 (ELECTRICAL AND ELECTRONIC FUNDAMENTALS LAB.)

EXPERIMENT NO. 3

NAME OF THE EXPERIMENT : VERIFICATION OF KVL

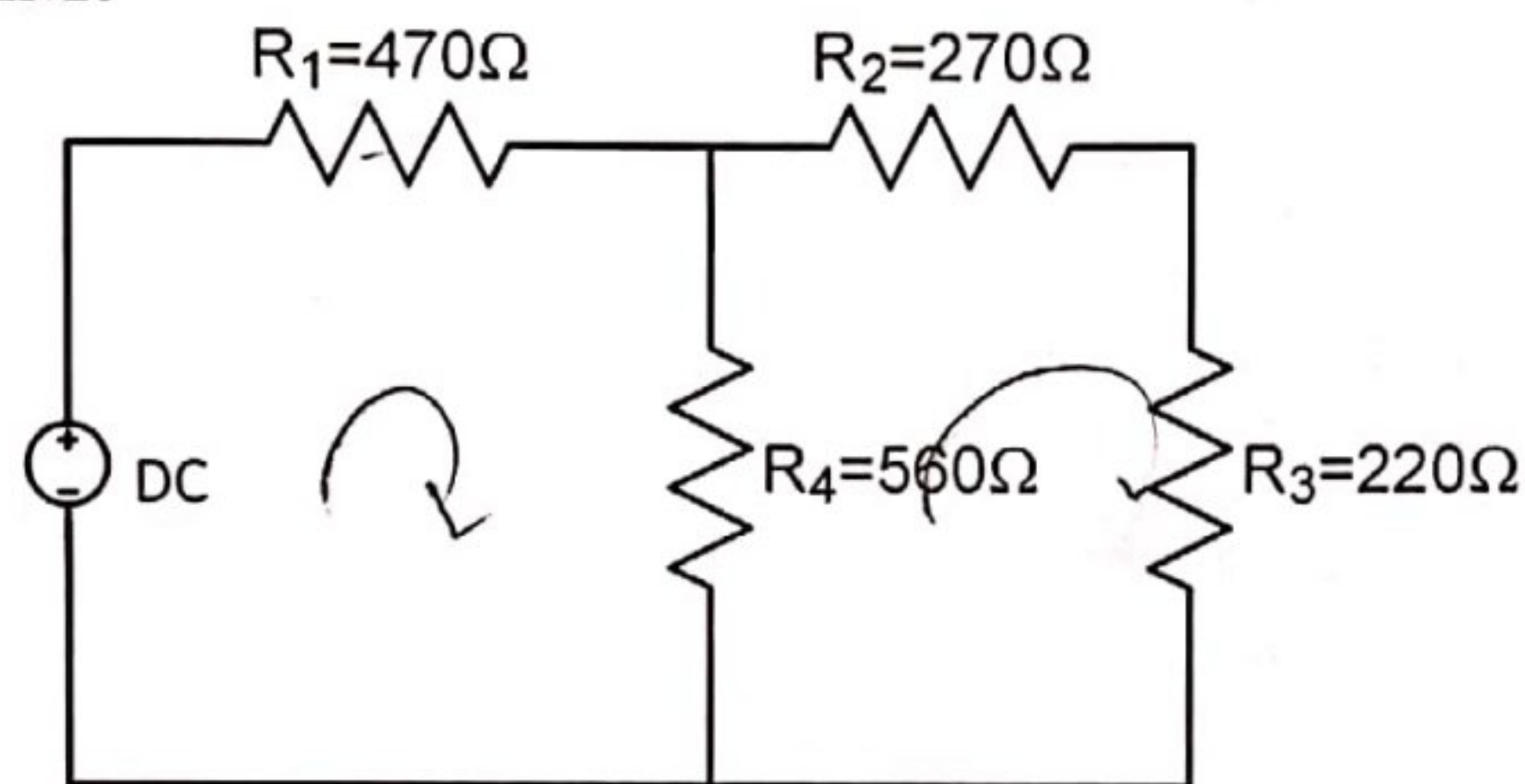
OBJECTIVE:

This experiment is intended to verify Kichhoff's Voltage Law (KVL) with the help of series-parallel circuits.

THEORY:

Around any closed circuit the algebraic sum of voltage rises equals the algebraic sum of the voltage drops.

CIRCUIT DIAGRAM:



$$V_{R4} = V_{R2} + V_{R3}$$

LABORATORY TASKS:

- i. Construct the above circuit.
- ii. Set source voltage at 5V.
- iii. Measure voltage across each resistor.
- iv. Note down data in a table as shown below.
- v. Interchange 470Ω and 560Ω.
- vi. Follow the instructions in (iii) to (iv) again.

DATA:

No of observation	Source voltage	Voltage across			
		R1	R2	R3	R4
1	5.07	3.231 ✓	0.993 ✓	0.821 ✓	1.815 ✓

CALCULATION:

Verify whether your data agree with KVL.

REPORT:

- (a) If any discrepancy is found then explain.
- (b) Can KVL be applied to open circuit?
- (c) Discussion.

for
10-01-23